§ 2.7

Find $y'(x)$.

11. $\cos(x^2y) - \sin y = x$

Sol. $(\cos(x^2y) - \sin y)' = (x)'$

$\Rightarrow -\sin(x^2y)\cdot(2xy + x^2y') - \cos y \cdot y' = 1$

$\Rightarrow -2xy\sin(x^2y) - x^2\sin(x^2y)\cdot y' - \cos y \cdot y' = 1$

$\Rightarrow -x^2\sin(x^2y)\cdot y' - \cos y \cdot y' = 1 + 2xy\sin(x^2y)$

$\Rightarrow (-x^2\sin(x^2y) - \cos y)\cdot y' = 1 + 2xy\sin(x^2y)$

$\Rightarrow y' = \dfrac{1 + 2xy\sin(x^2y)}{-x^2\sin(x^2y) - \cos y}$ #

16. $y\cos x^2 - \dfrac{3}{\sqrt{y^2+2}} = x^2 + 1$

$\dfrac{3}{\sqrt{y^2+2}} = 3(y^2+2)^{-1/2}$

Sol. $\left(y\cos x^2 - \dfrac{3}{\sqrt{y^2+2}}\right)' = (x^2+1)'$

$\Rightarrow \left(y'\cos x^2 - y\sin x^2\cdot 2x\right) + \dfrac{3}{2}(y^2+2)^{\frac{-3}{2}}\cdot 2y\cdot y' = 2x$

$\Rightarrow y'\left(\cos x^2 + 3y(y^2+3)^{\frac{-3}{2}}\right) = 2x + 2xy\sin x^2$

$\Rightarrow y' = \dfrac{2x(1 + y\sin x^2)}{\cos x^2 + 3y(y^2+3)^{-3/2}}$ #

Find an equation of the tangent line at the given point.

20. $x^3y^2 = -2xy - 3$ at $(-1, -3)$.

Sol. $(x^3y^2)' = (-2xy-3)' \Rightarrow 3x^2y^2 + 2x^3y \cdot y' = -2y - 2x \cdot y'$

$\Rightarrow 2x^3y \cdot y' + 2x \cdot y' = -2y - 3x^2y^2$

$\Rightarrow y' \cdot (2x^3y + 2x) = -2y - 3x^2y^2$

$\Rightarrow y' = \dfrac{-2y - 3x^2y^2}{2x^3y + 2x}$

$y'\big|_{x=-1, y=-3} = \dfrac{-2\cdot(-3) - 3\cdot 1\cdot 9}{2\cdot(-1)\cdot(-3) + 2\cdot(-1)} = \dfrac{6-27}{6+-2} = \dfrac{-21}{4}$

Hence the tangent line is $y + 3 = \dfrac{-21}{4}(x+1)$. #

Find $y''(x)$

24. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$

Sol. $(x^{\frac{2}{3}} + y^{\frac{2}{3}})' = (4)' \Rightarrow \dfrac{2}{3}x^{-\frac{1}{3}} + \dfrac{2}{3}y^{-\frac{1}{3}} \cdot y' = 0$

$\Rightarrow \dfrac{2}{3}y^{-\frac{1}{3}} \cdot y' = -\dfrac{2}{3}x^{-\frac{1}{3}}$

$\Rightarrow y' = \left(\dfrac{-x}{y}\right)^{\frac{1}{3}}$

$y'' = \dfrac{1}{3}\left(\dfrac{-x}{y}\right)^{-\frac{2}{3}} \cdot \left(\dfrac{-1}{y} + \dfrac{x}{y^2} \cdot y'\right)$

$= \dfrac{1}{3}\left(\dfrac{-x}{y}\right)^{-\frac{2}{3}} \cdot \left(\dfrac{-1}{y} + \dfrac{x}{y^2}\left(\dfrac{-x}{y}\right)^{\frac{1}{3}}\right)$ #

§2.8

In exercises 1–6, check the hypotheses of Rolle's Theorem and the Mean Value Theorem and find a value of c that makes the appropriate conclusion true. Illustrate the conclusion with a graph.

1. $f(x) = x^2 + 1, [-2, 2]$
2. $f(x) = x^2 + 1, [0, 2]$
3. $f(x) = x^3 + x^2, [0, 1]$
4. $f(x) = x^3 + x^2, [-1, 1]$
5. $f(x) = \sin x, [0, \pi/2]$
6. $f(x) = \sin x, [-\pi, 0]$

5. $f(x) = \sin x$, $[0, \frac{\pi}{2}]$

pf $f(0) = 0$, $f(\frac{\pi}{2}) = 1$

Since f is cont. on $[0, \frac{\pi}{2}]$ and diff. on $(0, \frac{\pi}{2})$, by the Mean Value Theorem, $\exists c \in (0, \frac{\pi}{2})$ s.t.

$$f'(c) = \frac{f(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2} - 0} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

$c \approx 0.88$

8. Prove that $x^3 + 4x - 3 = 0$ has exactly one solution.

pf Let $f(x) = x^3 + 4x - 3$, it is clear that f is cont and diff on $\mathbb{R}$.

Since $f(1) = 1 + 4 - 3 = 2$ and $f(0) = -3$, by the Intermediate Value Theorem, f has root on $(0, 1)$.

WLOG, if $0 < a < b < 1$ and $f(a) = f(b) = 0$, by the Rolle's Thm, $\exists c \in (a, b)$ s.t. $f'(c) = 0$. Given $f'(x) = 3x^2 + 4$,

$f'(c) = 0 \Leftrightarrow 3c^2 + 4 = 0$

$\Leftrightarrow c^2 = \frac{-4}{3} \Rightarrow c$ does not exist. ✗

Hence $x^3+4x-3=0$ has exactly one root. //

25. Prove that $|\sin a| < |a|$ for all $a \neq 0$ and use the result to show that the only solution to the equation $\sin x = x$ is $x = 0$. What happens if you try to find all intersections with a graphing calculator?

⇐ Since $|\sin a| < |a|$, $\forall x \neq 0$, only $x=0$ can let $\sin x = x = 0$. //

pf Let $f(x) = x - \sin x$

① $|\sin a| < |a|$, $\forall a \neq 0$

If $a \geq \frac{\pi}{2} > 1$, then $|\sin a| \leq 1 < a$

Thus $|\sin x| < a = |a|$, $\forall a \geq \frac{\pi}{2}$.

If $0 < a < \frac{\pi}{2}$, since f is cont. and diff. on $\mathbb{R}$, by the Mean Value Thm, $\exists c \in (0, a)$ s.t.

$$f'(c) = \frac{f(a) - f(0)}{a - 0} = \frac{a - \sin a}{a} = 1 - \frac{\sin a}{a}$$

Since $f'(x) = 1 - \cos x$, $f'(c) = 1 - \cos c = 1 - \frac{\sin a}{a}$

$$\Rightarrow \cos c = \frac{\sin a}{a}$$

$$\Rightarrow 1 > \cos c = \frac{\sin a}{a}$$

$$\Rightarrow \sin a < a$$

Hence $|\sin a| < |a|$, $\forall a > 0$.

If $a < 0$, let $b = -a > 0$, then $|\sin b| < |b|$

$\Rightarrow |\sin(-a)| = |-\sin a| < |-a| \Rightarrow |\sin a| < |a|$, $\forall a < 0$

Hence $|\sin a| < |a|$, $\forall a \neq 0$. //

§3.1

①

In exercises 1–6, find the linear approximation to $f(x)$ at $x = x_0$. Use the linear approximation to estimate the given number.

②

1. $f(x) = \sqrt{x}, x_0 = 1, \sqrt{1.2}$
2. $f(x) = (x+1)^{1/3}, x_0 = 0, \sqrt[3]{1.2}$
3. $f(x) = \sqrt{2x+9}, x_0 = 0, \sqrt{8.8}$
4. $f(x) = 2/x, x_0 = 1, \dfrac{2}{0.99}$
5. $f(x) = \sin 3x, x_0 = 0, \sin(0.3)$
6. $f(x) = \sin x, x_0 = \pi, \sin(3.0)$

① $L(x) = 0 + 3x = 3x$

③ $\sin(0.3) = f(0.1) \approx L(0.1) = 0.3$

3. $f(x) = \sqrt{2x+9}$, $x_0 = 0$, $\sqrt{8.8} = ?$

Sol. $f'(x) = \dfrac{2}{2\sqrt{2x+9}} = \dfrac{1}{\sqrt{2x+9}}$

① $L(x) = f(x_0) + f'(x_0)\cdot(x - x_0)$

$= \sqrt{9} + \frac{1}{\sqrt{9}}\cdot(x-0)$

$= 3 + \frac{1}{3}x$

② $\sqrt{8.8} = \sqrt{2\cdot(-0.1)+9} = f(-0.1)$

$\approx L(-0.1) = 3 + \frac{-0.1}{3}$

$= 3 - 0.033$

$= 2.967$ #

5. ① $L(x) = 3x$

② $\sin(0.3) \approx 0.3$

50. Show that Newton's method applied to $x^n - c = 0$ (where n and c are positive constants) produces the iterative scheme $x_{n+1} = \frac{1}{n}[(n-1)x_n + cx_n^{1-n}]$ for approximating $\sqrt[n]{c}$.

pf Since $x^n - C = 0 \Rightarrow x^n = C$

$\Rightarrow x = \sqrt[n]{C}$

$\Rightarrow \sqrt[n]{C}$ is a root of $x^n - C = 0$.

By Newton's Method, $x_{i+1} = x_i - \dfrac{f(x_i)}{f'(x_i)}$

$= x_i - \dfrac{x_i^n - C}{n x_i^{n-1}}$

$= \dfrac{1}{n}(n x_i - x_i + C x^{1-n})$ //

49. Show that Newton's method applied to $x^2 - c = 0$ (where $c > 0$ is some constant) produces the iterative scheme $x_{n+1} = \frac{1}{2}(x_n + c/x_n)$ for approximating $\sqrt{c}$. This scheme has been known for over 2000 years. To understand why it works, suppose that your initial guess (x_0) for $\sqrt{c}$ is a little too small. How would c/x_0 compare to $\sqrt{c}$? Explain why the average of x_0 and c/x_0 would give a better approximation to $\sqrt{c}$.

pf Let $f(x) = x^2 - C$. $\sqrt{C}$ is the root of f, and $f'(x) = 2x$

By Newton's Method, $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

$= x_n - \dfrac{x_n^2 - C}{2x_n}$

$= x_n - \dfrac{x_n}{2} + \dfrac{C}{2x_n}$

$= \dfrac{1}{2}\left(x_n + \dfrac{C}{x_n}\right)$. //